

Optics Letters

946-nm Nd:YAG digital-locked laser at 1.1×10^{-16} in 1 s and transfer-locked to a cryogenic silicon cavity

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Received 7 February 2019; accepted 1 March 2019; posted 7 March 2019 (Doc. ID 359609); published 29 March 2019

We present a Nd:YAG ultra-stable laser system operating at 946 nm and demonstrate a fractional frequency instability of 1.1×10^{-16} at 1 s by pre-stabilizing it to a 30-cm-long ULE cavity at room temperature. All key analog components have been replaced by FPGA-based digital electronics. To reach an instability below the 10^{-16} level, we transfer the stability of a 1542 nm laser stabilized to a cryogenic silicon cavity exhibiting a fractional frequency instability of 4×10^{-17} at 1 s to the laser at 946 nm. © 2019 Optical Society of America

<https://doi.org/10.1364/OL.44.001781>

Optical oscillators with high spectral purity have become key components of high-precision experiments, such as gravitational wave detection [1] and optical atomic clocks [2]. For the latter, short-term fractional frequency instabilities in the low 10^{-16} range or better are required for minimizing the Dick effect in optical lattice clocks [3], achieving long coherent interrogation times in ion clocks [2], or to benefit from the low quantum projection noise expected in multi-ions clocks [4–6]. Among all available technologies, ultra-stable lasers obtained by referencing a continuous wave (CW) laser to a high finesse Fabry–Perot cavity have been broadly preferred for their reliability and robustness. Many experiments could demonstrate short-term fractional frequency instabilities below 10^{-15} , but only a few have reached 1×10^{-16} or below to this date. The exceptionally low instabilities were either obtained at room temperature with very long cavities [7], which rely on a very carefully designed and optimized vibration-insensitive cavity support, or at cryogenic temperatures with technically more complex systems involving bulky cryocoolers [8]. Furthermore, the traditional electronics involved in the generation of the error signal and the stabilization itself are mostly analog, which have to be carefully designed and are not easy to tune in experiments, e.g., precise filter frequencies, phase adjustment of the Pound–Drever–Hall (PDH) signal, offset drift of the frequency mixer. In this Letter, we present a robust ultra-stable laser based on a solid-state Nd:YAG laser with high control bandwidth operating at 946 nm and stabilized to a 30 cm high finesse Fabry–Perot cavity placed on a simple adjustable vibration-insensitive support [9]. A similar approach has been followed recently [10]

with a 578 nm laser stabilized to two 30-cm-long ULE cavities, where a fractional frequency instability of 2×10^{-16} at 1 s was obtained but is still limited by the vibration sensitivities of the cavities. In this work, we demonstrate the first frequency stabilization via digital electronics at the fractional instability level of 1.1×10^{-16} at 1 s and discuss the relevant instability contributions. This laser will be used for high-precision spectroscopy of the $^1S_0 \leftrightarrow ^3P_0$ clock transition of $^{115}\text{In}^+$ in a multi-ion clock [4,6], for which it is frequency quadrupled to 236.5 nm. For reaching even better instabilities, we use a transfer-lock scheme to stabilize the 946 nm laser to a 1542 nm laser locked to a cryogenic silicon cavity with 4×10^{-17} fractional frequency instability from 0.8 s to ~ 30 s [8].

The laser used in this work is a home-built quasi-three level solid state Nd:YAG laser operating at 946 nm and is based on the design in Ref. [11]. The laser is depicted in Fig. 1. The laser

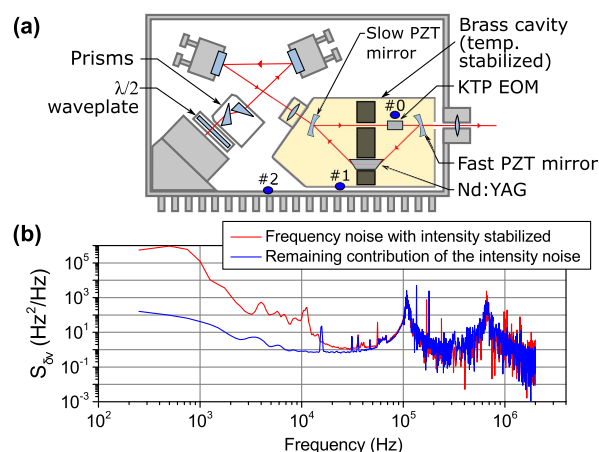


Fig. 1. (a) Design of the Nd:YAG laser at 946 nm. The blue dots represent the temperature sensors used for the active correction of the temperature gradients inside the laser cavity. (b) Frequency noise of the free-running laser. The red curve is obtained when the laser intensity is stabilized. The blue curve is the contribution of the remaining intensity noise. The peak at 110 kHz is due to a resonant relaxation oscillation of the Nd:YAG crystal [13]. The peak at 700 kHz is a servo bump from the intensity stabilization.

cavity mirrors are equipped respectively with a slow piezo-electric (PZT) module with control bandwidth ~ 100 Hz to extend the laser's frequency tuning range up to 2.5 GHz and a fast PZT module with control bandwidth ~ 15 kHz for fast corrections. We use a KTP crystal which combines two functions: intracavity electro-optic modulator (EOM) [12] and birefringent filter. The crystal axis is rotated at a small angle relative to the polarization of the radiation. The electro-optic properties of the crystal provide a high control bandwidth of 400 kHz. Its temperature can be used to tune the laser frequency by 120 GHz around 946 nm, while keeping an output power of ~ 200 mW. To have a very long operation time within the frequency tuning range of the EOM and PZT modules, the KTP crystal must be kept at a very stable temperature, which is difficult to achieve because of temperature gradients inside the laser cavity. We use temperature sensors placed at different positions on the laser head (Fig. 1) and find a correlation between the laser frequency and the temperature of the KTP crystal (sensor #0), the brass cavity enclosure (sensor #1) and the laser's heat sink (sensor #2). We use a digital controller to measure continuously the temperature of the three sensors and apply a feed-forward algorithm with weighted coefficients determined experimentally to correct the long-term fluctuations of the tuning range. After corrections the laser wavelength stays within 8 pm (2.5 GHz) over weeks. The free-running frequency noise of the laser is depicted in Fig. 1 and is about a factor 10 lower than the noise of laser diodes around this frequency. The measurement is performed on the side of a fringe of a Fabry-Perot cavity with a full width at half maximum of 10 MHz, which acts as a frequency-to-amplitude converter. To distinguish between frequency and amplitude noise, the laser intensity is stabilized with a double-pass acousto-optic modulator (AOM) with ~ 700 kHz bandwidth.

The laser is frequency stabilized to a high finesse Fabry-Perot cavity composed of a 30 cm ultra-low expansion (ULE) cuboid spacer [9] and fused silica mirrors with high reflection coatings at 946 nm. The calculated relative thermal noise limit of this cavity is 1×10^{-16} [14]. The cavity rests on a simple vibration insensitive support, which has previously been characterized with mirrors at 822 nm [9]. The measured acceleration sensitivities were in the low $10^{-11}/\text{ms}^{-2}$ for all directions. We exchanged the mirrors and placed them with tolerances better than 0.1 mm, so that deviations of the vibration sensitivities are expected to be below $4 \times 10^{-12}/\text{ms}^{-2}$. The system is mounted on a passive vibration isolation platform, which damps vibrations above 0.7 Hz. The cavity is placed inside a ~ 10 mm thick aluminium vacuum chamber, surrounded by an active thermal shield made of aluminium (see Fig. 2), which is temperature stabilized. Passive thermal insulation is provided by a gold-coated copper shield in vacuum, a polystyrene layer around the active thermal shield, and a simple wood box enclosure. We measure the frequency response of the cavity to a 0.5°C temperature step on the active shield and derive a time constant around 91 h. The zero-crossing of the cavity coefficient of thermal expansion is measured to be 40.4°C , very close to the simulated value of 39.6°C [15]. The laser frequency is stabilized to a cavity resonance via the PDH technique. The optical set-up is depicted in Fig. 2. The laser is placed on a different breadboard separated by a 10 m polarization maintaining (PM) fiber with active stabilization of the optical path. After a 60 dB Faraday isolator, the laser is phase modulated by

an EOM at 11.9 MHz. The residual amplitude modulation (RAM) of EOMs is one of the important instability contributions of PDH-stabilized lasers. It can be produced by interferences between the EOM facets [16] or from polarization rotation which converts to amplitude modulation after passing through polarizing optical elements [17]. EOMs based on crystals with Brewster-cut trapezoidal [18] or parallelogram [19] prism shape avoid interferences within the crystal and exploit its birefringence to separate ordinary and extra-ordinary components of the light in two beams (see Fig. 2), thus exhibiting very small intrinsic RAM. Here we use a home-built EOM based on a Brewster-cut trapezoidal prism KTP crystal for achieving a large separation of the two polarization components of the light. Using a photodiode placed directly behind the EOM, we rescale the amplitude fluctuations to voltage fluctuations of the PDH error signal and estimate a short-term passive fractional RAM below 2×10^{-6} . This is more than a factor 3 improvement over the ultra-low RAM obtained with parallelogram prism shape EOM [19] and is very close to the best results achieved via more complex active stabilization [17]. For our lock parameters, this contributes to a frequency instability below 2×10^{-17} at 4 s. The light reflected from the cavity is sent to a 50 MHz bandwidth photodiode. The transmitted light on resonance is sent to two 2.5 MHz bandwidth photodiodes for the intensity stabilization and its out-of-loop characterization.

The laser is frequency stabilized to the cavity by a digital controller using only the signal from the PDH photodiode. The frequency mixer, phase shifter, rf filters, and PID are implemented on an EP3C25 field-programmable gate array (FPGA) from Altera including 14 bit 65 Msps analog-to-digital converters and 125 Msps digital-to-analog converters. We program a quadrature lock-in amplifier [see Fig. 3(a)] at 11.9 MHz with two sines and cosines produced by a direct-digital synthesizer (DDS). The DDS signal is also used to drive the EOM. The demodulated PDH signal [20] after the mixer passes through a finite-impulse response filter to suppress harmonics of the modulation. Two digital PIDs are used in series for the stabilization of the laser to the cavity. We apply an original algorithm for the PID to have an adjustable limit K_{lim} of the integral gain K_i , which simplifies the optimization of the PID controllers. The operation principle of one PID is shown in Fig. 3(b). The maximum integrational gain is $K_i^{\text{max}} = K_i/(1 - K_{\text{lim}})$. The output of the digital lockbox is applied to the laser intracavity EOM, which provides the wide

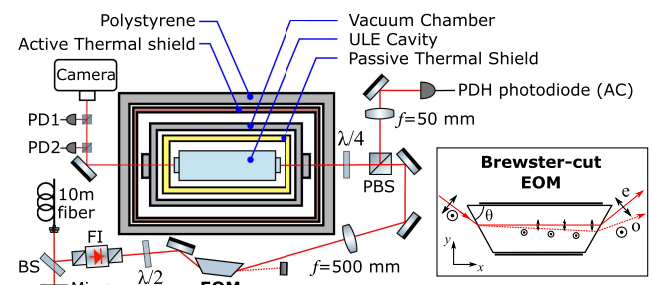


Fig. 2. Optical set-up for the PDH frequency stabilization to the cavity. BS, beam splitter; FI, Faraday isolator; PBS, polarizing beam splitter; PD1 and PD2, photodiodes in transmission. We use a trapezoidal EOM crystal cut at a Brewster angle θ to obtain a large separation of the ordinary ("o") and extra-ordinary ("e") polarization components of the light, thus achieving low RAM below 2×10^{-6} .

probably due to temperature inhomogeneities of the thermal shield, dominate the instability above 1 s. Figure 4(b) shows the measured instability depending on length stabilization of the fibered links. One can note that even the stabilization of the small 10 m fiber between the laser and the cavity is required to reach an instability of 10^{-16} .

This level of instability is suitable for interrogation of ~ 10 In⁺ ions in the multi-ion In⁺ clock [4,6]. Nevertheless, its theoretical quantum projection noise limit can be estimated at $7 \times 10^{-17} / \sqrt{\tau(\text{s})}$ for 100 In⁺ ions, which surpasses the performances of the interrogation laser [9]. We, therefore, implement the transfer oscillator technique presented previously to transfer the outstanding stability of the 1542 nm silicon cavity stabilized laser [8] to our 946 nm In⁺ clock laser. This technique was previously demonstrated in a stability transfer from the Si cavity laser to a Sr lattice clock laser [22] with an instability in the range of 1 to 3×10^{-16} at 1 s. This is also advantageous in reducing long-term instabilities due to temperature fluctuations of the cavity or the creeping of the ULE glass. Here, the transfer beat at 55 MHz is used in a control loop with bandwidth of ~ 160 Hz to eliminate the relative fluctuations and, thus, transfer the stability by means of an AOM frequency shifter as actuator for the frequency of the 946 nm In⁺ clock interrogation field [see Fig. 5(a)]. The fractional frequency instability of the transfer beat computed from the individual beat notes with the comb is depicted in Fig. 5(b) when the 946 nm laser is only locked to the ULE cavity and when the transfer scheme is active (in-loop measurement). The in-loop instability shows that we correctly stabilize to the silicon cavity and that the additional noise due to the control loop is still compatible with an instability of 7×10^{-18} at 1 s. We are currently limited by the signal-to-noise ratio of the beats with the comb (~ 80 dBc/Hz), which can be observed as a $\tau^{-1.5}$ slope in Figs. 4 and 5(b), and by excess noise in the transfer electronics. The actual 946 nm laser instability after transfer-lock may currently be limited by uncompensated path length fluctuations between the different branches of the comb, which can lead to instability contributions as high as $2 \times 10^{-16} / \tau(\text{s})$ [22]. In the future this will be replaced by a common branch of the comb, where instabilities well below 10^{-17} can be achieved above 1 s [23].

In summary, we have presented a laser system at 946 nm based on a home-built Nd:YAG laser with a high control bandwidth, which is stabilized to a 30 cm ULE Fabry–Perot cavity using a Brewster-cut EOM with short-term passive intrinsic RAM below 2×10^{-6} . To stabilize the laser to the cavity, we developed a digital controller replacing all key components of the PDH stabilization scheme. The controller offers several advantages over classical analog schemes, such as easy configuration of the PID parameters, precision control of the phase, and the mixer is free of offset drift. We demonstrate with our system a fractional frequency instability of 1.1×10^{-16} at 1 s with long-term performance dominated by non-linear drifts probably due to temperature inhomogeneities of the outer aluminium shield, which can be improved by better thermal contacts and a more homogeneous heat load. Our approach allows for realizing robust and easy-to-operate ultra-stable lasers at room temperature at the 10^{-16} level. To reach ultimate performances, we further stabilize the 946 nm laser to a 1542 nm laser locked to a cryogenic silicon cavity exhibiting a fractional frequency instability of 4×10^{-17} at 1 s. In the future, using crystalline coatings [24] or nanostructured mirrors [25] may

allow us to reach instabilities in the 10^{-17} range with simple room temperature systems.

Funding. Russian Foundation for Basic Research (RFBR) (16-52-12045); Deutsche Forschungsgemeinschaft (DFG) (ME 3648/1-3, CRC 1227 (DQ-mat), project B03); Horizon 2020 Framework Programme (H2020) (Project 15SIB03 OC18 from the EMPIR programme), cofinanced by the Participating States.

Acknowledgment. The authors thank T. Legero and U. Sterr for providing the silicon cavity signal, M. Skvortsov for designing and developing the low-RAM Brewster-cut EOM, and S. A. King for careful reading of this Letter.

REFERENCES

1. R. X. Adhikari, *Rev. Mod. Phys.* **86**, 121 (2014).
2. A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, and P. O. Schmidt, *Rev. Mod. Phys.* **87**, 637 (2015).
3. N. Hinkley, J. A. Sherman, N. B. Phillips, M. Schioppa, N. D. Lemke, K. Beloy, M. Pizzocaro, C. W. Oates, and A. D. Ludlow, *Science* **341**, 1215 (2013).
4. N. Herschbach, K. Pyka, J. Keller, and T. E. Mehlstäubler, *Appl. Phys. B* **107**, 891 (2012).
5. K. Arnold, E. Hagiye, E. Paez, B. C. Hui Lee, J. Bollinger, and M. Barrett, *Phys. Rev. A* **92**, 032108 (2015).
6. J. Keller, T. Burgermeister, D. Kalincev, A. Didier, A. P. Kulosa, T. Nordmann, J. Kiethe, and T. E. Mehlstäubler, *Phys. Rev. A* **99**, 013405 (2019).
7. S. Häfner, S. Falke, C. Grebing, S. Vogt, T. Legero, M. Merimaa, C. Lisdat, and U. Sterr, *Opt. Lett.* **40**, 2112 (2015).
8. D. G. Matei, T. Legero, S. Häfner, C. Grebing, R. Weyrich, W. Zhang, L. Sonderhouse, J. M. Robinson, J. Ye, F. Riehle, and U. Sterr, *Phys. Rev. Lett.* **118**, 263202 (2017).
9. J. Keller, S. Ignatovich, S. A. Webster, and T. E. Mehlstäubler, *Appl. Phys. B* **116**, 203 (2014).
10. L. Jin, Y. Jiang, Y. Yao, H. Yu, Z. Bi, and L. Ma, *Opt. Express* **26**, 18699 (2018).
11. M. V. Okhapkin, M. N. Skvortsov, A. M. Belkin, and S. N. Bagayev, *Opt. Commun.* **194**, 207 (2001).
12. V. A. Rusov, V. A. Serebryakov, A. B. Kaplun, and A. V. Gorchakov, *J. Opt. Technol.* **76**, 325 (2009).
13. C. C. Harb, T. C. Ralph, E. H. Huntington, D. E. McClelland, H.-A. Bachor, and I. Freitag, *J. Opt. Soc. Am. B* **14**, 2936 (1997).
14. T. Kessler, T. Legero, and U. Sterr, *J. Opt. Soc. Am. B* **29**, 178 (2012).
15. T. Legero, T. Kessler, and U. Sterr, *J. Opt. Soc. Am. B* **27**, 914 (2010).
16. E. A. Whittaker, M. Gehrtz, and G. C. Bjorklund, *J. Opt. Soc. Am. B* **2**, 1320 (1985).
17. W. Zhang, M. J. Martin, C. Benko, J. L. Hall, J. Ye, C. Hagemann, T. Legero, U. Sterr, F. Riehle, G. D. Cole, and M. Aspelmeyer, *Opt. Lett.* **39**, 1980 (2014).
18. K. L. Dooley, M. A. Arain, D. Feldbaum, V. V. Frolov, M. Heintze, D. Hoak, E. A. Khazanov, A. Lucianetti, R. M. Martin, G. Mueller, O. Palashov, V. Quetschke, D. H. Reitze, R. L. Savage, D. B. Tanner, L. F. Williams, and W. Wu, *Rev. Sci. Instr.* **83**, 033109 (2012).
19. Z. Tai, L. Yan, Y. Zhang, W. Guo, S. Zhang, and H. Jiang, *Opt. Lett.* **41**, 5584 (2016).
20. R. W. P. Drever, J. L. Hall, F. V. Kowalski, J. Hough, G. M. Ford, A. J. Munley, and H. Ward, *Appl. Phys. B* **31**, 97 (1983).
21. J. Stenger, H. Schnatz, C. Tamm, and H. R. Telle, *Phys. Rev. Lett.* **88**, 073601 (2002).
22. C. Hagemann, C. Grebing, T. Kessler, S. Falke, N. Lemke, C. Lisdat, H. Schnatz, F. Riehle, and U. Sterr, *IEEE Trans. Instr. Meas.* **62**, 1556 (2013).
23. H. Leopardi, J. Davila-Rodriguez, F. Quinlan, J. Olson, J. A. Sherman, S. A. Diddams, and T. M. Fortier, *Optica* **4**, 879 (2017).
24. G. D. Cole, W. Zhang, B. J. Bjork, D. Follman, P. Heu, C. Deutsch, L. Sonderhouse, J. Robinson, C. Franz, A. Alexandrovski, M. Notcutt, O. H. Heckl, J. Ye, and M. Aspelmeyer, *Optica* **3**, 647 (2016).
25. J. Dickmann and S. Kroker, *Phys. Rev. D* **98**, 082003 (2018).